

4 (short) The 3-pin arch shown in Fig. 4 has a right-hand side whose shape is given by:

$$y = h \left(\frac{x}{L} \right)^3$$

The left-hand side of the arch is formed of two straight sections, rigidly connected at A. The horizontal section is loaded with a uniform downwards vertical load of w per unit width.

- (a) Calculate the bending moment at point A. [5]
- (b) Calculate the location of the maximum bending moment in the right-hand side of the arch. [5]

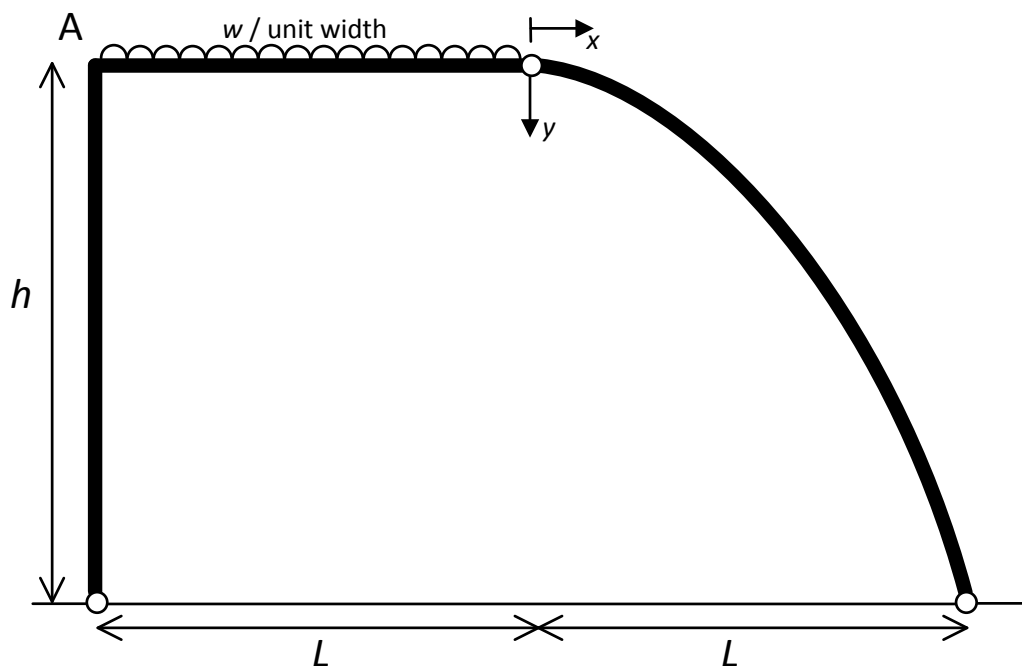


Fig. 4